

EXP.NO-5	Write a Servlet to demonstrate the difference between HTTP GET and POST methods by creating a form and handling requests accordingly.

AIM:

To create a web form to accept user input (name and age) and use a Servlet to process and display the data on the browser.

ALGORITHM:**1.Input:**

The user enters a product name in the web form and submits it using either GET or POST method

2.Create the Web Form:

Design an HTML page with two forms:

- One using GET method
- One using POST method

Both forms contain an input field for product name and a submit button. Set the form action to the Servlet URL.

3.Create the Servlet:

Create a Servlet class that extends HttpServlet.
Use @WebServlet annotation to map the servlet.
Override both doGet() and doPost() methods.

4.Handle GET Request:

In doGet(), retrieve the input using:
`request.getParameter("name")`
Display the product name along with a message indicating GET method.

5. Handle POST Request:

In doPost(), retrieve the input similarly.
- Display the product name along with a message indicating POST method.

6. Apply Styling:

Use an external CSS file to maintain consistent UI for both GET and POST outputs.

7. Deploy and Run:

Run the project on Apache Tomcat server.
Open the HTML page in browser and test both methods.

8. Observe Difference:

In GET method, data is visible in the URL.
In POST method, data is not visible in the URL.

CODE:

getpost.html

```
<!DOCTYPE html>

<html>

<head>

    <title>TechScope - Compare Products</title>

    <style>

        body {

            font-family: Arial;

            background: #0b1220;

            color: white;

            display: flex;

            justify-content: center;

            align-items: center;

            height: 100vh;

        }
```

```
.card {  
    background: #020617;  
    padding: 30px;  
    border-radius: 12px;  
    width: 350px;  
    box-shadow: 0 10px 30px rgba(0,0,0,0.5);  
    text-align: center;  
}  
  
h1 {  
    color: #38bdf8;  
}  
  
input {  
    width: 100%;padding: 10px;  
    margin: 10px 0 20px;border-radius: 8px;  
    border: none;  
}  
  
button {  
    width: 100%;padding: 10px;margin-bottom: 10px;  
    border: none;border-radius: 8px;  
    font-weight: bold;cursor: pointer;  
}  
  
.get-btn {  
    background: #22c55e;  
}  
  
.post-btn {  
    background: #3b82f6;
```

```
    }

    button:hover {

        opacity: 0.9;

    }

</style>
</head>
<body>

<div class="card">

    <h1>TechScope</h1>
    <p>Search & Compare Products</p>
    <!-- GET -->
    <form action="gp" method="get">
        <input type="text" name="name" placeholder="Enter product (GET)">
        <button class="get-btn">Search using GET</button>
    </form>
    <!-- POST -->
    <form action="gp" method="post">
        <input type="text" name="name" placeholder="Enter product (POST)">
        <button class="post-btn">Search using POST</button>
    </form>

</div>

</body>
</html>
```

GetPostservlet.java

```
package com.techscope;
```

```
import java.io.*;
import jakarta.servlet.*;
import jakarta.servlet.http.*;
import jakarta.servlet.annotation.WebServlet;

@WebServlet("/gp")
public class GetPostServlet extends HttpServlet {

    protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        response.setContentType("text/html");

        String name = request.getParameter("name");
        String method = "GET";
        String color = "#22c55e";

        PrintWriter out = response.getWriter();

        out.println("<html>");
        out.println("<head>");
        out.println("<link rel='stylesheet' href='style.css'>");
        out.println("</head>");
        out.println("<body>");
        out.println("<div class='card'>");
        out.println("<h1 class='title'>TechScope    </h1>");
        out.println("<h2 class='get'>GET Method Used</h2>");
        out.println("<p>Product: <b>" + name + "</b></p>");
        out.println("</div>");
        out.println("</body></html>");
```

```
}
```

```
protected void doPost(HttpServletRequest request, HttpServletResponse response)
```

```
    throws ServletException, IOException {
```

```
    response.setContentType("text/html");
```

```
    String name = request.getParameter("name");
```

```
    String method = "POST";
```

```
    String color = "#3b82f6";
```

```
    PrintWriter out = response.getWriter();
```

```
    out.println("<html>");
```

```
    out.println("<head>");
```

```
    out.println("<link rel='stylesheet' href='style.css'>");
```

```
    out.println("</head>");
```

```
    out.println("<body>");
```

```
    out.println("<div class='card'>");
```

```
    out.println("<h1 class='title'>TechScope </h1>");
```

```
    out.println("<h2 class='post'>POST Method Used</h2>");
```

```
    out.println("<p>Product: <b>" + name + "</b></p>");
```

```
    out.println("</div>");
```

```
    out.println("</body></html>");
```

```
}
```

```
}
```

style.css

```
body {
```

```
    font-family: Arial;
```

```
    background: #0b1220;
```

```
    color: white;
```

```
display: flex;
justify-content: center;
align-items: center;
height: 100vh;
}

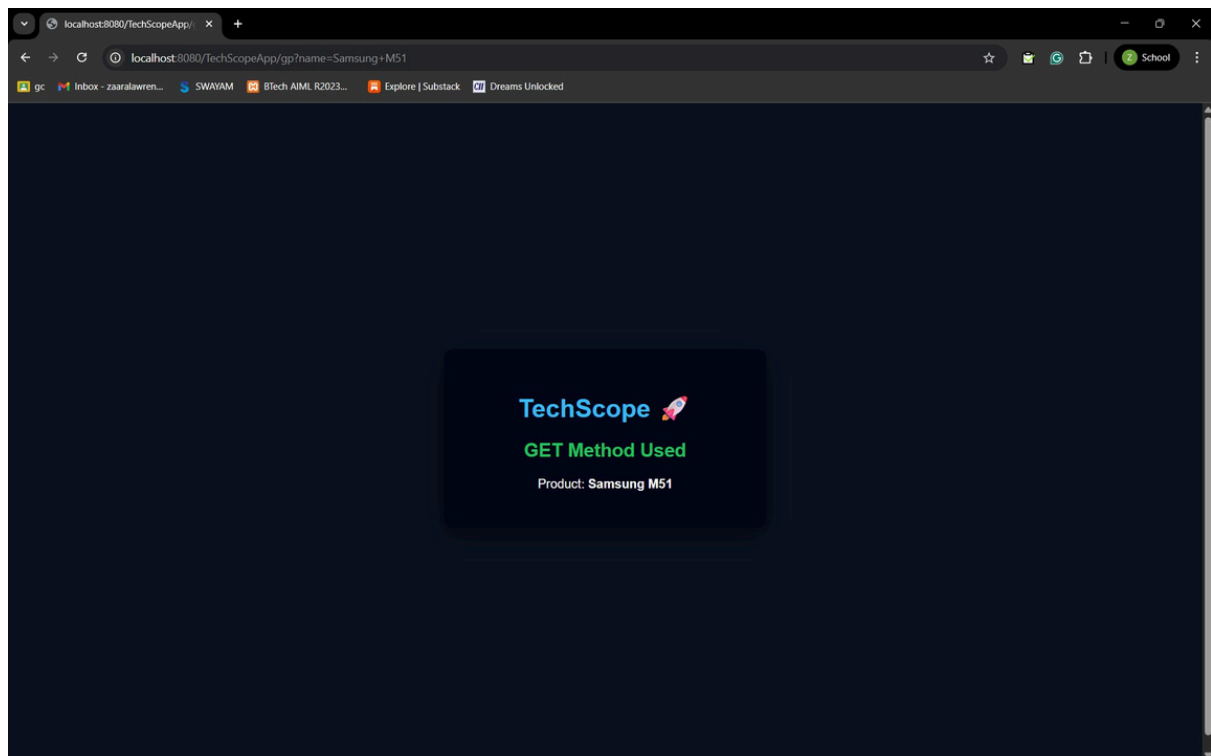
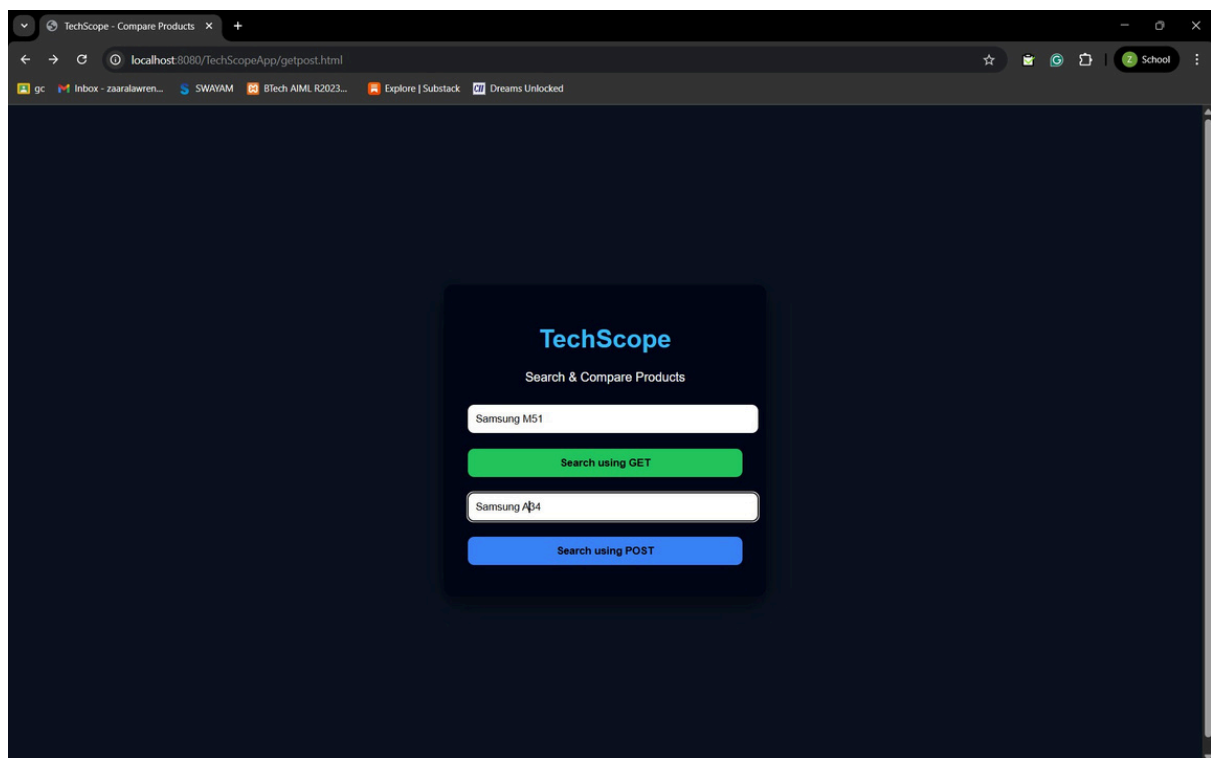
.card {
background: #020617;
padding: 30px;
border-radius: 12px;
width: 350px;
text-align: center;
box-shadow: 0 10px 30px rgba(0,0,0,0.5);
}

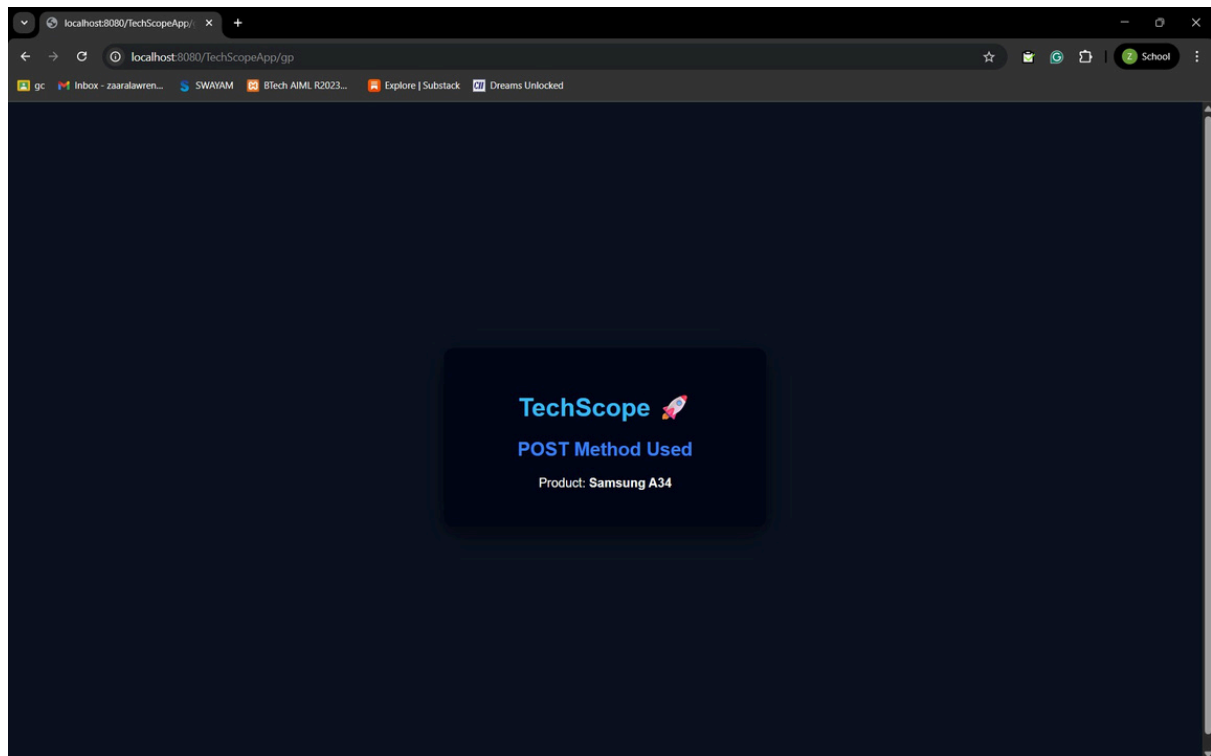
.title {
color: #38bdf8;
}

.get {
color: #22c55e;
}

.post {
color: #3b82f6;
}
```

OUTPUT:





RESULT:

Thus, the difference between HTTP GET and POST methods was successfully demonstrated using a Servlet. The GET method appends data in the URL, while the POST method sends data securely in the request body.